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Chapter 15 Abstract In this chapter, we present a new optimization algorithm called the imperialist competitive algorithm (ICA), which is inspired by the human socio-political evolution process. We first describe the general knowledge of the imperialism in the United States.Sect. 15.1. Then, the fundamentals and performance of ICA are introduced in Sect. 1.15.2. Finally, 1st section, 15.3, summarises this chapter. 15.1 Introduction to the 'empire'In politics and history, in order to explain the extending control over weaker people or areas, the term "empire" is appeared. One of the famous examples is the reign of the Greco-Roman Empire from 27 BC to 476. The most powerful person is usually called an imperialist and different imperialists will extend their authority and control of one state or people over another through the form of competition. Inspired by this human socio-political evolution process, a newly developed evolutionary algo- rithm called Imperialist competitive algorithm (ICA) is proposed by Atashpaz-e-Gargari and Lucas (2007). 15.1.1 Imperialism Imperialism was greatly influenced by an economic theory known as mercantilism, which inspired the government to extend their power and the rule beyond its own boundaries (Atashpaz-Gargari and Lucas 2007). In its initial forms, due to the limited supply of wealth, counties focused on building their empire with natural resources, such as gold and silver. However, nowadays, the new imperialism focused on the new technological advances and developments. The ultimate goal is to increase the number of their colonies and spread their empires over the world. 203 B. Xing and W.-J. Gao, Innovative Computational Intelligence, Intelligent Systems Reference Library 62, DOI: 10.1007/978-3-319-03404-1_15, Springer International Publishing Switzerland 2014.

204.15.2 Imperialist Competitive Algorithm204.14.2.1 Fundamentals of Imperialist competitive Algorithm (ICA)204.13.1.1 Imperialist competitive algorithm (ICAI) was originally proposed by Atashpaz-e-Gargari and Lucas (2007). In ICA, the countries can be viewed as population individuals and basically divided into two groups based on their power, i.e., colonies and imperialists. Also, one empire is formed by one imperialist with its colonies. Furthermore, two operators called assimilation and revolution and one strategy called imperialistic competition are the main building blocks that employed in ICA. The implementation procedures are described as below:• Initializing phase: 1.1. Preparation of initial populations. Each solution (i.e., country) that in form of a country-to-country-an array can be defined via Eq. 15.1 (Atashpaz-Gargari and Lucas 2007):country ¼ p1; p2; . . Ance, 1.0.1.2.3.1, N1.0,000.

PNvar = 1½ ð15:1Þ where pis represent different variables which based on various socio-political characteristics (such as culture, language, and economical policy), and Nvar denotes the total number of the characteristics (i.e., n-dimension of the prob-lems) to be optimized. 2. Creating the cost function. In order to evaluate the cost of countries, the cost-of-function can be defined via Eq. 15.2 (Atashpaz-Gargari and Lucas 2007): . . . pcos t ¼ f country; p1; p2; p3; p4; p5; p6; p7; p8; p9; . . Ance, 1.0.1.2.3.1, N1.0,000. PNvar:1Þ:1:1ð15:2Þ;1:3. Initializing the empires: In general, the initial size of populations (Npop)involves two types of countries [i.e., colony (Ncol) and imperialist (Nimp)] which together form the empires. To form the initial empires proportionally, the normalized cost of an imperialist is defined via Eq. 15.3 (Atashpaz-Gargari and M.D. DeLucas 2007): 1-2NCn ¼ cn maxi ci f g; ð15:3Þ where cn is the cost of nth imperialist, NCn denotes its normalized cost. Normally, two methods can be used to divide colonies among imperialists: (1) from the imperialists' point of view which based on the power of each imperialist; (2) from the colonies' point of view which based on the relationship with the imperialist (i.e., the colonies should be possessed by the imperialist according to the power). Both methods are given via Eqs. 15.4 and 15.5, respectively (Atash-paz-Gargari and Lucas 2007).

205 ILCSPowern ¼1 NOCn; NNCn i¼1 NCi ð15:4Þ; Ncol ð16:5ÞNOCn ¾1 Nimp; Nimp ¼2 Nimp, Ncol and Nimp represent the number of all colonies and imperialists, respectively, and NOC nimp is the initial number of colonies of nth empire. • Moving phase: 1.1. Assimilation strategy: According to this strategy, all colonies will move toward an imperialist with x units, and all imperialist with y units will move towards an imperialist. This strategy was implemented by the imperialist. x units via Eq. 15.6 (Atashpaz-Gargari and Lucas (2007):x U 0; b d 0; x U 1Þ; ð15:6Þ where x is a random variable with uniform distribution, b is a number greater than 1, and d is the distance between a colony and an imperialist; 1, 2. Revolution strategy, 1. According to this strategy, a random amount of deviation from the directionof movementto the direction of movement is incorporated via Eq. 15.7 (Atashpaz-Gargari and Lucas 2007):Eq.15.7:h U c; C 15:7 where h is a random variable with uniform distribution, and c is a parameter that adjusts the deviation from the original direction. • • • • Based on the cost function, when the new position of a colony is better than that of the corresponding imperialist, the imperialist and the colony change their positions and the new location with a lower cost becomes the imperialist. • Imperialistic competition phase: 1-1. Calculating the total power of an empire: It is influenced by the power of a non-imperialist country and the colonies of an imperialist country via Eq. 15.8 (Atashpaz-eq.15.8):TCn ¼ Cost imperialistn ¾ Cost imperialistÞ þ n mean Cost colonies of empires. 15:8 where TCn is the total cost of the nth empire, and n is a positive number which is considered to be less than 1. 1, 2. Imperialistic competition strategy, 1. According to this strategy, all empires try to take control of the colonies of

other empires and control them. To be able to do so, you need to be able to do so.

Once the Empire has modelled this strategy, the weakest colonies of the weakest empires will be chose to competition among all other empires in order to possess this colony. Based on TCN, the normalized total cost is simply obtained via Eq. 15.9, 15.5, 15,5.5%-9.5% TCN maxi TCN ð15.6% where NTCN represents the total normalized cost of the empire. Having NTCN, the possession probability of each empire is evaluated via Eq. 15.5.10.10 (Atashpaz-Prospectpn.3.3) MacBookNTCN i.e., 1 NTCI, haircut, haircutty15.5,3.2,2.3,3,4,4-3,6-4,6,7-7,7,8-6,8,7.4,746,632,732,846,814,832,946,914,932 • Convergence phase: At the end, all the colonies will be under the control of the most powerful empire, which means all the colonies have the same positions and costs and will be controlled by an imperialist with the same position and cost as themselves. In other words, there are no differences not only between colonies, but also between colonies and imperialists. Taking into account the key phases described above, the steps of implementing ICA can be summarized as follows (Atashpaz-Gargari and Lucas 2007): • Step 1: Defining the optimization problem. Left Step 2.1.2. Generating initial empires by pick some random points on the function. • Step 3: Move the colonies towards imperialist states in different directions (i.e., through the process of re-assimilation). • Step 4: Random changes occur in the characteristics of some countries (i.e., the result of a counter-revolution). • Step 5: Position exchange between a colony and an imperialist. Step 6: Calculate the total cost of all empires. • Step 7: Use imperialistic competition and pick the weakest colony from the next-weakest empire. • Step 8: Eliminate the powerless empires. 1st Step 9.0.1. Check if maximum iteration is reached, go to Step 3 for new beginning. If a specified termination criteria is satisfied, stop and return the best solution.

207.15.2.2 Performance of ICA In order to show how the ICA performs, four minimization problems are tested in this study:Atashpaz-Gargari and Lucas (2007). Compared with genetic algorithm (GA) and particle swarm optimization (PSO), the results showed that ICA is capable of reaching the global minimum. 15.3 Conclusions and RecommendationsIn this chapter, an optimization algorithm motivated by one of the socio-politically models (i.e., imperialistic competition) is introduced. The term "countries" are designed as the initial populations and categorized into colony and imperialist states, respectively. Both colonies and imperialists together form the empires. The basic idea behind ICA is to lead the search process toward the powerful imperialist or the optimum points based on their 'power", i.e., the weakest empires will lose their colonies until there will be no colony in that. The objective of ICA is to find an ideal world in which there is no difference not only between colonies but also between colonies and imperialists. Although it is a newly introduced computational intelligence method, we have witnessed the following rapid spreading of ICA: First, several enhanced versions of ICA can be found in the literature as out-of-print code below:• Bacterial foraging optimization based on ICA (Acharya et al. 2010). Left Chaotic improved ICA (Talatahari et al. 2006). 2012a, b). • Constrained genetic algorithm based ICA (Acharya et al. 2010). Fast ICA (Acharya et al. 2010). Diagnostic Hybrid artificial neural network and ICA (Taghavifar et al. 2013). • Hybrid evolutionary algorithm and ICA (Ramezani et al. 2012). • Modified ICA (Niknam et al. 2011). Project Multiobjective ICA. (Bijami et al. 2011; Mohammadi et al. 2011). • • • • Quad countries algorithm (Soltani-Sarvestani et al. 2006). 2012). Second, the ICA has also been successfully applied to a variety of optimization problems, as listed below, namely Artificial neural network training (Moadi et al. 2011). • Assembly line balancing (Bagher et al. 2011). • Data mining (Ghanavati et al. 2011; Niknam et al. 2011). • Facility location optimization (Mohammadi et al. 2011; Moadi et al. 2011). • Power system optimization (Nejad and Jahani 2011; Bijami et al. 2011). Product mix-outsourcing problem (Nazari-Shirkouhi et al. 2010). • Scheduling optimization (Forouharfard and Zandieh 2010; Ayough et al. 2012; Lian et al. 2012; Kayvanfar and Zandieh 2012).

208.• Soil compaction prediction (Taghavifar et al. 2013). • Structure design optimization (Talatahari et al. 2012b). Interested readers please refer to them as a starting point for further exploration and exploitation of ICA.
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